

The Bay a Shogunate Could Not Defend: Yokosuka's Iron Works and Naval Port

A scattering of fishing hamlets on a drowned, hilly coast became the shogunate's last great construction project, built to the design of a French engineer by the finance minister it later executed as a rebel. Reclaimed ground, a naval district, an earthquake and an occupying navy have kept remaking the shoreline ever since; the US Navy holds much of it still.



横須賀（神奈川県横須賀市）

In this dossier

1. A history of the district (7 chapters, about 1400 words)
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Written in English from Japanese library sources — chiefly books digitised by the National Diet Library — and official records. Superscript numbers point to the numbered sources at the end, with page frames for the digitised books.

A scattered hamlet on a drowned coast

Yokosuka's harbour is a drowned valley, and the Geospatial Information Authority's landform survey still reads the ground around it as mostly former water, slope and reclaimed fill rather than solid natural terrace. Before the shipyard that made the city's name, an 1917 study of the district describes only a scattering of fishing hamlets here: twelve households on the western shore of Katsuriki Point, and some ninety-six in total across the inlet's edge, with travellers between Uraga and Edo crossing by ferry near the old Suwa Shrine. Otaki-cho, now the densest commercial street in the city, was then, the same source records, no more than two or three thatched huts at the foot of a hill, reachable on foot along the shore only at low tide.¹⁹

Perry's ships, and the shipyard that followed

The bay's transformation began offshore. In 1853 American warships entered Uraga Bay, a short distance south of this study area at Kurihama, breaking, in the words of the 1917 study, "the two-hundred-year dream" of the shogunate's closed coast; when the Americans returned that autumn, the shogunate dropped its old ban on building large ships and let individual domains try. Mito's lord had a three-masted Western-style sailing ship built at Uraga; another was built at Ishikawajima in Edo Bay; the shogunate and the Saga domain each bought a steamer from abroad. None of it added up to a shipbuilding industry, and an 1857 ironworks the shogunate opened at Nagasaki, using ten hired Dutch engineers, proved too remote and too small to build a proper warship.¹

The answer was a purpose-built yard inside Edo Bay, and the man who pushed it through was Oguri Tadamasa, better known by his court title Kozuke-no-suke, at the time the shogunate's finance commissioner. In 1864 he and the inspector Kurimoto Sehei opened talks with the French minister Léon Roches, who recommended a French engineer then working in Shanghai, François Léonce Verny. Oguri's own first idea, an 1934 memorial pamphlet on his career records, was modest: a \$40,000 repair yard using machinery already donated by the Saga domain. Roches's enthusiasm expanded it into a four-year, \$2.4 million project, and construction began with a groundbreaking ceremony on 27 September 1865, to Verny's design — the direct origin, by the shogunate's own account, of what is now the Yokosuka Naval Arsenal.¹²

The magistrate who built a navy and died a rebel

Oguri did not live to see his shipyard finished. Born in 1827 into a hereditary bannerman family with a 2,500-koku stipend, he had already served on the 1860 mission to Washington that exchanged ratifications of Japan's treaty with the United States, and held the finance commissioner's post three separate times before Yokosuka. When the shogunate's forces were defeated at Toba-Fushimi in January 1868, Oguri argued for continued resistance and was, according to the 1934 pamphlet's account, personally dismissed by the shogun himself — the only instance, it states, of a shogunal retainer being dismissed by direct order of the shogun in the whole of the Tokugawa period. He withdrew to his own fief at Kota village in Kozuke province, was accused of plotting insurrection by the new imperial army advancing from the north, and was captured and beheaded beside the Karasu River on 6 April 1868, at the age of forty-two. The shipyard he had founded outlived him by more than a century and a half, and now forms the core of a modern naval base.²

Reclaiming a naval city

Around the growing arsenal, the shoreline itself had to be manufactured. The 1917 study traces the sequence in detail: Shioiri was created in 1869 by levelling a hill into a reed-choked wetland; the flat approach to what became Yokosuka Station was built in 1871 by filling part of Hemmi village's coast and cutting down another hill to form Minato-cho; ten formal town names, among them Motomachi, Suwa and Otaki, were only fixed in May 1875, replacing paddies and forest that until then had no names at all. Private reclamation followed through the late 1870s — a group led by Ogawa Shigechika filled ground east of Inaoka in 1878, and a Kamoi innkeeper named Takahashi Katsushichi reclaimed the shore east of Otaki in 1879 to create Wakamatsu-cho, cutting a new road toward Fukada village. In 1880 the whole of Tomarisato was bought up for the

shipyard's own expansion. Yokosuka absorbed Hemmi village in 1889 and reached city status on 15 February 1907. The Imperial Navy formalised the site's strategic role in between: in 1884, the first of what became four Japanese naval districts was raised here, ahead of Kure and Sasebo in 1889 and Maizuru in 1901, and it was under French technical guidance that the yard built Japan's first stone dry dock and a two-ton steam hammer, before Japanese-trained engineers went on to build the country's third dry dock unaided.¹⁴

1923: a city built on fill

Yokosuka sat close enough to the Great Kanto earthquake's Sagami Bay epicentre that the 1932 official record of the disaster calls its shaking exceptionally severe. Much of the city collapsed outright, fires broke out in several quarters, and a huge landslide at the Minato-cho cliff buried the road under what the record describes as tens of thousands of tsubo of earth, killing residents caught beneath it; a rumour that more than 150 schoolgirls on an outing had been crushed there spread for months before debris clearance proved it false. City Hall's own building collapsed, killing or injuring several staff, and for months afterward the city government worked from tents pitched in the grounds of the Naval District headquarters, itself nearly levelled but spared by fire; the Naval District's commander-in-chief was made martial-law commander within three days.³

The 1932 record is unusually direct about why the damage fell where it did: the city sat mostly on reclaimed ground, cut hillside or valley fill rather than natural soil, and it was precisely this soft, loose ground that failed hardest, with coastal revetments collapsing wholesale into the sea and the naval port's own north-facing frontage nearly destroyed. The pattern the 1932 surveyors traced by eye — collapse concentrated on filled and cut ground, comparative safety on the old natural slopes — is the same one the modern land-classification maps of this district still show today.³⁹

1945: one navy replaces another

Surrender did not empty the naval base; it changed its flag. Film held in the US National Archives, and catalogued by Japan's own National Showa Memorial Museum, shows American Marines coming ashore at "Green Beach" on Yokosuka's southern waterfront on 30 August 1945, days before the formal surrender, moving through the city's streets to secure the former Japanese base while disarmament continued at Kurihama into early September. The US Navy has held part of the old arsenal ever since; today's Fleet Activities Yokosuka reports something like 24,500 military and civilian personnel supporting the US Navy's forward-deployed forces in the Western Pacific, on grounds that overlap the yard Verny built.⁵⁸

Reading the ground today

The battleship Mikasa, built in England in 1902 and Admiral Togo Heihachiro's flagship in the Russo-Japanese War, was designated a memorial ship in 1926 and is preserved today, by its own trust's account, as the world's oldest surviving steel battleship; postwar neglect nearly destroyed it before a preservation campaign restored it by the early 1960s, and it now sits landlocked in concrete at the head of Mikasa Park, at the harbour's edge. A few streets inland, Dobuita-dori still carries the name of a drainage ditch that once ran down its centre: the naval arsenal, the shopping street's own association records, supplied the thick iron plates once used to cover it so people and carts could pass, and the plates gave the street its lasting name even after both ditch and iron were long since removed.⁶⁷

Between the two, the town's oldest bones are easiest to find in what does not sit on fill: the old shrine ground, the natural slopes the 1923 record judged safest, and the line of the original shore now buried under Otaki-cho and Minato-cho. Almost everywhere else — the naval district's grounds, the shipyard itself, the flat approach to the station — is reclaimed or cut ground built in the four decades after Oguri's 1865 groundbreaking, a sequence still visible in the sharp edge where the Miura Peninsula's hills give way to the flat, filled shelf the modern port and base now occupy.¹⁹

Timeline

1853	American warships under Perry enter Uruga Bay, just south of this district, ending the shogunate's closed-coast policy. ¹
1857	The shogunate's first ironworks, at Nagasaki, proves too small and remote to build large warships. ¹
1864	Finance commissioner Oguri Kozuke-no-suke opens talks with French minister Léon Roches on a new shipyard at Yokosuka. ^{1,2}
27 September 1865	Groundbreaking ceremony for the Yokosuka Ironworks, designed by the French engineer Léonce Verny. ²
6 April 1868	Oguri Kozuke-no-suke, dismissed by the shogun and accused of insurrection by the new imperial army, is beheaded at Kota, Kozuke province. ²
1869-1880	Shioiri, Minato-cho, Ogawa-cho and Wakamatsu-cho are created by successive land reclamation along the shore. ¹
1884	Yokosuka Naval District is established, the first of Japan's four naval districts. ⁴
15 February 1907	Yokosuka is granted city status. ¹
1 September 1923	The Great Kanto earthquake devastates Yokosuka, with the worst damage on reclaimed and cut ground. ³
1926	The battleship Mikasa is designated a memorial ship. ⁶
30 August 1945	US Marines land at Green Beach, Yokosuka, beginning the American Navy's continuous presence at the base. ⁵
1958-1961	A preservation campaign restores the neglected Mikasa; it is rededicated in 1961. ⁶

Walking notes

Places to stand and look, in walking order. Straight-line distances only; check today's access and opening before you go.

- 1

Verny Park and the former Ironworks waterfront ヱェルニー公園. Faces the US Navy base across the water, close to where Léonce Verny's Yokosuka Ironworks broke ground in 1865; named for the engineer himself.²
- 2

Former Yokosuka Naval District grounds 旧横須賀鎮守府. Site of the naval district established in 1884, nearly destroyed in the 1923 earthquake but pressed into service as a tent city for both naval and municipal government afterward.^{3,4}
- 3

Dobuita-dori どぶ板通り. Named for the iron plates the naval arsenal once supplied to cover a drainage ditch running down the street; long an America-facing shopping street and the birthplace of the souvenir sukajan jacket.⁷
- 4

Mikasa Park and the battleship Mikasa 三笠公園・記念艦みかさ. The 1902-built flagship of the Russo-Japanese War, designated a memorial ship in 1926, nearly lost to postwar neglect before restoration returned it to display by the early 1960s.⁶
- 5

Otaki-cho and Minato-cho 大滝町・港町. Now the city's central shopping streets, both built on land reclaimed between 1869 and the 1880s from what an 1917 study describes as a shoreline of a few thatched huts.¹

The same ground, 1945–1950 and 2021



1945–1950 aerial photograph (US military photography) — GSI Tiles *ort_USA10*, cropped and annotated by Japan Time Atlas. North is up; the frame is identical in both images.



2021 annual orthophoto — GSI Tiles *nendophoto2021*, cropped and annotated by Japan Time Atlas. Capture dates within a mosaic can differ from place to place.

How to compare. Anchor on features that rarely move — river banks, rail lines, shrine and temple grounds, main roads — then look for what changed around them. Differences in tone come from film, camera and season, not from the land itself.

The decades in between

Every further GSI series that covers the same frame, oldest first. Each series is a compilation: photographs within one series were taken on different dates, which is why seams and changes of tone can appear inside a frame.



1961–1969 · GSI Tiles *ort_old10*, processed



1974–1978 · GSI Tiles *gazo1*, processed



1979–1983 · GSI Tiles *gazo2*, processed



1987–1990 · GSI Tiles *gazo4*, processed

Rivers, shorelines and terraces



Natural landforms as classified by GSI, over a recent aerial photograph in gray. Dark blue outlines mark former river channels.

Former water area 31%	Mountain and hill slopes 29%
Water 24%	Floodplain and coastal plain 8.3%
Foothill deposits 7.1%	Sandbars and dunes under 0.5%

What the main landforms mean

Former water area (31%). Land shown by old maps and aerial photographs to have been sea, lake, pond, reservoir or river, later turned into land. Soft ground: very strong liquefaction tendency where filled, strong where drained. Storm surge near the coast.

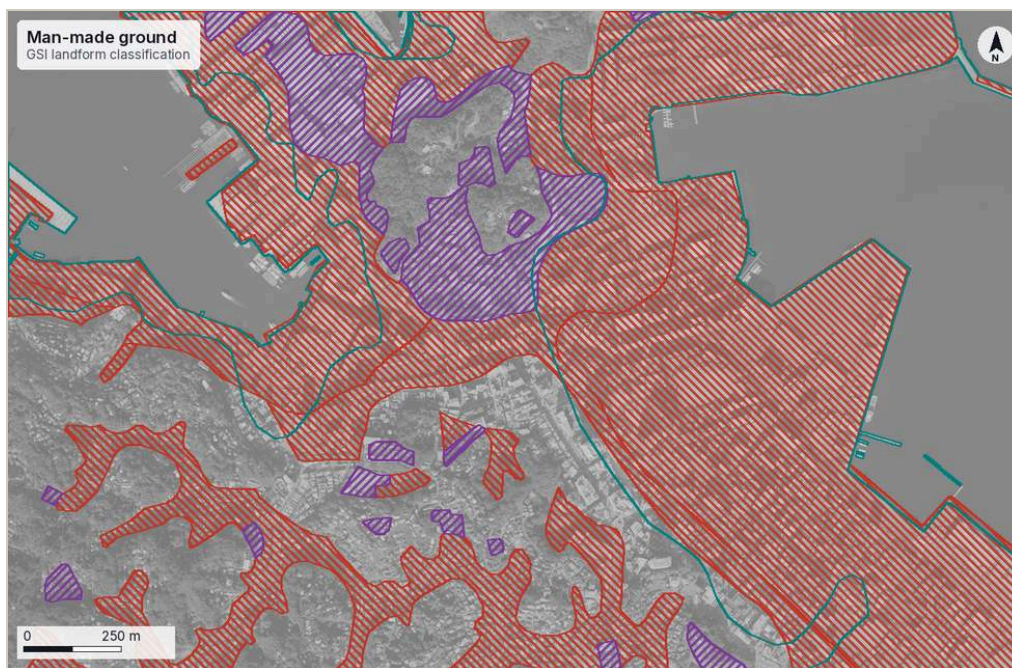
Mountain and hill slopes (29%). Ridges, valleys and fairly steep ground, including terrace-scarp slopes and volcanic ground. Heavy rain or earthquakes can trigger slope failures, debris flows and landslides.

Floodplain and coastal plain (8.3%). Low, flat ground of little relief, built from sand and mud left by floods, or former seabed that emerged. River flooding. Ground is softer nearer the coast and shakes somewhat more; liquefaction tendency is strong on coastal plains and somewhat strong on floodplains. Storm surge near the coast.

Foothill deposits (7.1%). Gentler ground below mountains and scarps, built from debris brought down by slope failures and debris flows. Debris flows are possible in heavy rain; the ground is unstable and slopes may fail in earthquakes.

Descriptions translated and condensed from GSI's landform-classification legend. They describe tendencies of each landform type, not the condition of any individual plot.

Fill, reclamation and the fields beneath the town



Hatching: man-made ground as classified by GSI. Teal outlines: former sea, river or pond. Dark blue outlines: former river channels.



GSI “Meiji-era lowland” data, traced from topographic maps of the 1880s–1900s; GSI warns that positions can be off by a considerable distance, especially in the Kanto and Kinki regions. Only low, wet or open ground was traced: towns, dry fields and forest of the period are left blank.

Man-made ground

-  Filled and reclaimed land 45%
-  Cut ground 6.1%

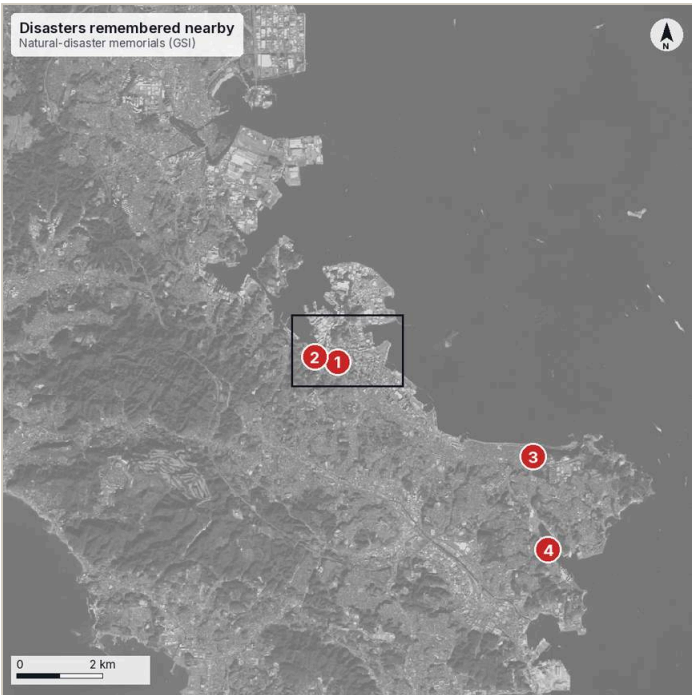
Meiji-era lowland

-  River, pond or sea 46%
-  Tidal flat or beach 2.9%
-  Rice paddy under 0.5%

Filled and reclaimed land (45%). Ground raised with fill, land made by dumping soil into the sea, rivers or ponds, or valleys filled in. May flood if not raised enough; on hills, rain or earthquakes can make the fill fail; in lowlands the liquefaction tendency is very strong, especially on filled sea, lake or river beds.

Cut ground (6.1%). Ground made by cutting into slopes of mountains or terrace edges. Depending on the cut slope, heavy rain or earthquakes may bring it down; the ground itself is generally firm.

Memorials to natural disasters nearby



Numbered memorials; the black rectangle is the study area of this brief. Frame 16 km across.
Fewer than three memorials stand within 3 km of the study area, so the nearest ones within 8 km are listed.

	Memorial	Disaster	Erected	Distance
1	Memorial to the refuge at Suwa Shrine after the 1923 Great Kanto Earthquake 大震災避難記念碑	Great Kanto Earthquake (1 Sep 1923) · earthquake	1925	0.3 km
2	Memorial to the Yokosuka victims of the 1923 Great Kanto Earthquake 大震災遭難者供養塔	Great Kanto Earthquake (1 Sep 1923) · landslide or debris flow / earthquake	1929	0.8 km
3	Memorial to the 1923 earthquake dead who washed ashore at Mabori 大震災歿死者之霊	Great Kanto Earthquake (1 Sep 1923) · earthquake / other	1930	5.0 km
4	Memorial tower to the victims of the 1923 Great Kanto Earthquake in Uraga 関東大震災慰霊塔	Great Kanto Earthquake (1 Sep 1923) · landslide or debris flow / earthquake	1972	6.6 km

What the memorials record

1. **Memorial to the refuge at Suwa Shrine after the 1923 Great Kanto Earthquake.** After the Great Kanto Earthquake of 1 September 1923, the number of people sheltering at Suwa Shrine reached 400 by 5 September, and some lived at the shrine for a while. The monument was put up by the evacuees themselves.
2. **Memorial to the Yokosuka victims of the 1923 Great Kanto Earthquake.** The Great Kanto Earthquake of 1 September 1923 was a disaster without precedent. Yokosuka did not escape: there were scenes of devastation everywhere, and 517 people in the city died. The memorial was moved to this site in 1936; nearby are a stone listing the victims' names and a Jizo statue dedicated to them.
3. **Memorial to the 1923 earthquake dead who washed ashore at Mabori.** In the days and weeks after the Great Kanto Earthquake of 1 September 1923, around 10 bodies, thought to be victims from Tokyo and Yokohama, washed ashore on the Mabori coast of Yokosuka and were buried here. The memorial was erected in 1930, seven years after the disaster.
4. **Memorial tower to the victims of the 1923 Great Kanto Earthquake in Uraga.** The Great Kanto Earthquake of 1 September 1923 caused slope collapses and fires in Uraga. In Nishi-Uraga, Mount Atago collapsed, burying 74 houses and about 100 people, and at Dokkuhara a slide buried 7 passers-by alive. The former town of Uraga alone lost 202 people. Rebuilt after the war, the tower honours victims of the earthquake and of shipwrecks.

Source: GSI Natural Disaster Memorials (自然災害伝承碑). English names and notes are Japan Time Atlas translations and summaries of GSI's Japanese descriptions; the memorials' own inscriptions may say more. "Erected" is the year GSI records for the monument.

Sources

1. 綾部虎治郎 (Ayabe Torajiro), 横須賀研究：附・三浦半島誌 [Yokosuka studies, with a gazetteer of the Miura Peninsula], 1917; NDL frames 27, 28, 29. <https://dl.ndl.go.jp/pid/958183/1/27> (open online (NDL Digital Collections, no login)). Used for: Pre-1865 fishing hamlets; the Perry-to-Nagasaki-ironworks narrative; the sequence of land reclamation 1869-1907 and city status in 1907.
2. 石原戒造編 (Ishihara Kaizo, ed.), 幕末の英傑小栗上野介を偲ぶ：横須賀海軍工廠創設の由来 [Remembering the late-shogunate hero Oguri Kozuke-no-suke: the origin of the Yokosuka Naval Arsenal], 1934; NDL frames 12, 13, 15, 21, 22. <https://dl.ndl.go.jp/pid/1098400/1/13> (open online (NDL Digital Collections, no login)). Used for: Oguri's negotiations with Léon Roches and Verny, the 1865 groundbreaking, the cost expansion from Oguri's original modest plan, and Oguri's biography and 1868 execution.
3. 横須賀市震災誌刊行会編 (Yokosuka City Earthquake Record Publishing Committee, ed.), 横須賀市震災誌：附・復興誌 [Yokosuka City's earthquake record, with a reconstruction record], 1932; NDL frames 39, 40, 41, 46. <https://dl.ndl.go.jp/pid/1466332/1/39> (open online (NDL Digital Collections, no login)). Used for: The 1923 earthquake's effects on the city, the Minato-cho landslide, the tent-city administration, and the explicit link between damage and reclaimed/cut ground.
4. 鎮守府 横須賀・呉・佐世保・舞鶴 [The naval districts: Yokosuka, Kure, Sasebo, Maizuru], 文化庁 日本遺産ポータルサイト (Agency for Cultural Affairs, Japan Heritage portal). <https://japan-heritage.bunka.go.jp/ja/stories/story035/>, checked 2026-09-28. Used for: Yokosuka Naval District's 1884 founding as the first of Japan's four naval districts, and the yard's early dry-dock and steam-hammer technology.
5. 本州の横須賀への最初の占領軍の到着 [The first occupation forces' arrival at Yokosuka, Honshu], 昭和館デジタルアーカイブ (National Showa Memorial Museum digital archive). <https://search.showakan.go.jp/search/video/detail.php?id=90148851>, checked 2026-09-28. Used for: The 30 August 1945 Marine landing at Green Beach, Yokosuka, and disarmament operations continuing to early September.
6. 世界三大記念艦「三笠」 [One of the world's three great memorial ships: Mikasa], 公益財団法人三笠保存会 (Battleship Mikasa Preservation Society). <https://www.kinenkan-mikasa.or.jp/>, checked 2026-09-28. Used for: Mikasa's 1902 construction, its role as Togo Heihachiro's flagship, and its 1926 designation as a memorial ship.
7. *All about* どぶ板 [All about Dobuita], ドブ板通り商店街 (Dobuita-dori Shopping Street Association). <https://dobuita-st.com/all-about>, checked 2026-09-28. Used for: The street's name origin in the naval arsenal's iron ditch-covers, and its postwar development as an America-facing shopping street.
8. *Commander Fleet Activities Yokosuka*, Military Times Installation Guide. https://installationguide.militarytimes.com/installation_service/commander-fleet-activities-yokosuka/, checked 2026-09-28. Used for: Current personnel figures (about 24,500) and mission statement for the US Navy base, used after official US Navy sites (history.navy.mil, cnrj.cnmc.navy.mil) returned server errors this session.
9. 地形分類 (自然地形・人工地形) ベクトルタイル [Landform classification (natural and artificial) vector tiles], 国土地理院 (Geospatial Information Authority of Japan). https://github.com/gsi-cyberjapan/experimental_landformclassification, checked 2026-09-28. Used for: Classification of the study area as predominantly former water, slope and artificial fill.
10. 地理院タイル 空中写真 (米軍撮影ほか) 及び全国最新写真 (シームレス) [GSI Tiles: aerial photographs (including US forces) and current seamless photographs], 国土地理院 (Geospatial Information Authority of Japan). <https://maps.gsi.go.jp/development/ichiran.html>, checked 2026-09-28. Used for: Basis for comparing historical and present-day aerial photography of the study area.

Further reading

- 横須賀市史 (横須賀市史編纂委員会編, 1957) — Yokosuka City's own full postwar history, with far more detail on the base, the 1923 quake and the interwar city than this dossier can cover. <https://dl.ndl.go.jp/pid/3025011> (NDL individual digital transmission (registered users))
- 横須賀百年史 (横須賀百年史編さん委員会編, 1965) — A centennial history of the city, covering the arsenal, the naval district and the postwar base transition. <https://dl.ndl.go.jp/pid/2995616> (NDL individual digital transmission (registered users))
- 横須賀市と基地：基地対策のあゆみと国有財産の転用 (横須賀市渉外部編, various years 1976-2000) — The city's own long-running series on managing the US base's presence and former naval land, useful for the postwar and present-day base relationship. <https://dl.ndl.go.jp/pid/13895275> (NDL premises only)
- 旧・横須賀鎮守府庁舎&ドライドック：横須賀に眠る文化遺産 (長浜つぐお編, 1998) — A dedicated study of the surviving Naval District headquarters building and dry docks. <https://dl.ndl.go.jp/pid/14087286> (NDL premises only)

Where a Deep Research request would go next

- **US Navy real-estate and requisition records for Yokosuka, 1945-1952** — US National Archives (NARA): SCAP/GHQ real estate division records; US Naval Forces Far East records. Search the NARA catalog for Yokosuka base requisition and occupation-period property files.
- **Detailed casualty and reconstruction figures for the 1923 earthquake in Yokosuka specifically** — Yokosuka City Museum (横須賀市自然・人文博物館) and the Kanagawa Prefectural Archives. Consult their disaster-history collections and any surviving ward-level damage maps.

- **Household-level records for the pre-1865 fishing hamlets absorbed into the Ironworks site** — Yokosuka City Museum and the National Archives of Japan (国立公文書館). Search for surviving Edo-period village registers (人別帳) for the Katsuriki and Oppama coastal hamlets.
- **US Navy personnel and family-housing history at Yokosuka through the Cold War** — US Naval History and Heritage Command and the Naval War College's archives. Contact the Command Historian's office directly, since history.navy.mil's own installation pages were unreachable this session.

Caveats

- Perry's 1853 landing itself took place at Kurihama, about 5 km south of this district's study centre, not within the study area; it is described here only as regional context for the Ironworks that followed.
- The battleship Mikasa's postwar neglect (a dance hall and aquarium once installed aboard) and its 1958-1961 restoration are well attested in Japanese press accounts, but could not be confirmed this session on the preservation society's own site, which gave only the ship's construction, 1905 service and 1926 designation directly.
- Current US Navy base population and mission figures come from a military-affairs trade publication rather than the Navy's own site, because history.navy.mil and cnrj.cnrc.navy.mil both returned server errors (503 and 403) to this session; the figures should be treated as approximate and rechecked against an official Navy source when available.
- Casualty numbers for the 1923 earthquake in Yokosuka specifically (as opposed to the wider Kanto region) are not stated as a single figure in the sources read this session; the dossier instead describes the qualitative pattern of destruction the 1932 record itself emphasises.

Digitised books are cited, not reproduced: the page frames let you open the same page in the NDL Digital Collections. Aerial photographs, landform and memorial data: GSI, processed by Japan Time Atlas. This experimental edition is a draft for testing the product and has not had independent fact-checking.